

Rational-Flux Anisotropic Harper Dynamics: Chambers Phase Collapse and the Algebraic Bloch-Band Atlas

HCS-C371 / HEN-O355

4 September 2026

Abstract

For every reduced rational magnetic flux and every positive hopping anisotropy, we fix one magnetic-cell phase convention and derive the exact Chambers characteristic identity. The proof exposes why the vertical transfer trace has only three Laurent modes and fixes both phase signs.

1 Magnetic owner and phase convention

Let $(p, q) = 1$, $q \geq 3$, and $\lambda > 0$. On $\ell^2(\mathbb{Z}^2)$ consider the bounded self-adjoint operator

$$\begin{aligned} (\mathbf{H}_{p/q, \lambda} \Psi)_{m, n} &= \Psi_{m+1, n} + \Psi_{m-1, n} \\ &\quad + \lambda e^{2\pi i p m / q} \Psi_{m, n+1} + \lambda e^{-2\pi i p m / q} \Psi_{m, n-1}. \end{aligned} \quad (1)$$

Fourier transformation in the second coordinate and magnetic Bloch reduction give the q -dimensional fiber

$$(\mathbf{H}(k_x, k_y)u)_m = u_{m+1} + u_{m-1} + 2\lambda \cos\left(k_y + \frac{2\pi p m}{q}\right) u_m, \quad u_{m+q} = e^{i q k_x} u_m. \quad (2)$$

Thus $X = qk_x$ is the *total* horizontal boundary phase. This distinction is essential: throughout,

$$D_{p/q, \lambda}(E; k_x, k_y) := \det(EI_q - \mathbf{H}_{p/q, \lambda}(k_x, k_y)). \quad (3)$$

This manuscript's first revision is the **Chambers phase-collapse owner**.

2 The Chambers polynomial

Put $\zeta = e^{2\pi i / q}$, $y = e^{i k_y}$, and

$$A_m(E, y) = \begin{pmatrix} E - \lambda(y\zeta^{pm} + y^{-1}\zeta^{-pm}) & -1 \\ 1 & 0 \end{pmatrix}, \quad M_q = A_{q-1} \cdots A_0. \quad (4)$$

Theorem 1 (Anisotropic Chambers identity). *There is a unique real monic polynomial $\mathcal{P}_{p/q, \lambda}$ of degree q such that*

$$D_{p/q, \lambda}(E; k_x, k_y) = \mathcal{P}_{p/q, \lambda}(E) - 2\cos(qk_x) - 2\lambda^q \cos(qk_y) \quad (5)$$

for every real E, k_x, k_y .

Proof. Since $\det A_m = 1$, the Floquet multipliers of M_q are $e^{\pm i q k_x}$. Both sides below are monic of degree q and have the same roots, hence

$$D_{p/q, \lambda} = \text{tr } M_q - 2\cos(qk_x). \quad (6)$$

Replacing y by $\zeta^p y$ cyclically permutes the potential sequence in (4). The monodromy changes by cyclic conjugacy, so its trace is invariant. Because p is invertible modulo q , a Laurent polynomial of y with degrees between $-q$ and q and this invariance can use only the modes $-q, 0, q$.

The coefficient of y^q has one contribution: select $-\lambda y \zeta^{pm}$ in every upper-left entry. It equals

$$(-\lambda)^q \zeta^{pq(q-1)/2} = -\lambda^q. \quad (7)$$

Indeed, odd q makes $q-1$ even, while even q forces p odd. Complex conjugation gives the same coefficient for y^{-q} . The remaining mode is independent of both phases, real, and monic; call it $\mathcal{P}_{p/q, \lambda}(E)$. Equations (6) and (7) prove (5). Monicity gives uniqueness. $\square$

The theorem is convention-complete: using the boundary character X in place of k_x rewrites the horizontal term as $-2\cos X$, not $-2\cos(qX)$. This is precisely the convention tested by the executable fibers.

References

- [1] P. G. Harper, *Single band motion of conduction electrons in a uniform magnetic field*, Proc. Phys. Soc. A 68 (1955), 874–878.
- [2] W. G. Chambers, *Linear-network model for magnetic breakdown in two dimensions*, Phys. Rev. 140 (1965), A135–A143.
- [3] D. R. Hofstadter, *Energy levels and wave functions of Bloch electrons in rational and irrational magnetic fields*, Phys. Rev. B 14 (1976), 2239–2249.
- [4] J. Avron, P. H. M. van Mouche, and B. Simon, *On the measure of the spectrum for the almost Mathieu operator*, Comm. Math. Phys. 132 (1990), 103–118.
- [5] M. P. Lamoureux and J. A. Mingo, *On the characteristic polynomial of the almost Mathieu operator*, Proc. Amer. Math. Soc. 135 (2007), 3205–3215. <https://doi.org/10.1090/S0002-9939-07-08830-2>.
- [6] S. Jitomirskaya, L. Konstantinov, and I. Krasovsky, *On the spectrum of critical almost Mathieu operators in the rational case*, arXiv:2007.01005.